MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM:** 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 objective questions (1 mark each).
2. Section B has 2 Short Answer questions (2 marks each). Section C has 2 questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (6 marks). Show all steps clearly.
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show all steps and calculations clearly.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each) [4 Marks]

1. A solid metallic sphere of radius r is melted and cast into the shape of a solid cone of height r. The radius of the base of the cone is: **[1]**

(a) (a) r (b) $2r$ (c) $3r$ (d) $4r$

2. A metallic solid cone is melted to form a solid cylinder of equal radius. If the height of the cylinder is 6 cm, then the height of the cone was: **[1]**

(a) (a) 10 cm (b) 12 cm (c) 18 cm (d) 24 cm

3. A rectangular sheet of paper 40 cm $\times$ 22 cm is rolled along its breadth to form a hollow cylinder of height 40 cm. The radius of the cylinder is: **[1]**

(a) (a) 3.5 cm (b) 7 cm (c) $\frac{80}{7}$ cm (d) 5 cm

4. **Assertion (A):** Two cubes each of surface area 96 cm^2 are joined end-to-end to form a cuboid. The volume of the resulting cuboid is 128 cm^3.
Reason (R): If n cubes each of surface area S are joined end-to-end, the volume of the resulting cuboid is $n\left(\frac{S}{6}\right)^{3/2}$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

5. A solid is in the shape of a cone surmounted on a hemisphere, the radius of each of them being 3.5 cm and the total height of the solid being 9.5 cm. Find the volume of the solid. **[2]**

6. The largest possible sphere is carved out of a wooden solid cube of side 7 cm. Find the volume of the wood left. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

7. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy. **[3]**

8. A hemispherical depression is cut out from one face of a cubical wooden block of edge 21 cm, such that the diameter of the hemisphere is equal to the edge of the cube. Determine the volume and total surface area of the remaining block. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (6 Marks) [6 Marks]

9. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm.

(i) Calculate the total volume of the iron pole. (Use $\pi = 3.14$)

(ii) Find the mass of the pole in kilograms, given that 1 cm^3 of iron has approximately 8 g mass.

[6]

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